

Market Regime Discovery

Unsupervised Clustering of Historical Market States

CMPT 455 — Introduction to Machine Learning · Winter 2026

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Shiller S&P 500 Dataset · 1,556 samples · 1881–2023

1. Introduction

Financial markets have historically alternated between distinct macroeconomic environments — periods of irrational exuberance, prolonged stagnation, high-rate regimes, and sharp crashes. Identifying these regimes from historical data, without imposing a predefined taxonomy, is a canonical unsupervised learning problem. This project applies clustering to 142 years of monthly S&P 500 macroeconomic data (Shiller 1881–2023) to discover latent market regimes driven purely by the data.

Problem type: Unsupervised learning — clustering. No labels are provided; structure must emerge entirely from 8 engineered macro features across 1,556 observations.

Why clustering? A supervised UP/DOWN frame imposes a look-ahead bias and is too coarse to capture the qualitative diversity of market environments. Clustering asks a richer question: what fundamentally different economic worlds has the market lived in? The answer has practical applications in regime-aware portfolio construction, risk management, and semi-supervised forecasting pipelines.

Challenges: The data is high-dimensional (8 features), exhibits non-stationarity over 142 years, and has no ground-truth labels. Macro time series are heteroskedastic and contain extreme events (Great Depression, 1970s inflation). Defining a meaningful k requires both quantitative criteria and domain interpretability.

Contributions: (1) Novel 8-feature engineering pipeline from raw Shiller data; (2) systematic k -selection using three complementary metrics; (3) comparison of K-Means, Agglomerative, and DBSCAN; (4) cluster profiling with economic interpretation; (5) regime transition matrix revealing market stickiness.

2. Methods

2.1 Data

The dataset is the Shiller S&P 500 monthly dataset, spanning January 1871 to February 2026 (1,710 raw rows). After removing rows with zero-encoded missing values — concentrated in the earliest pre-1881 period — the working dataset is 1,556 clean monthly observations (January 1881 – June 2023). Raw columns include nominal price, dividends, earnings, CPI, long interest rate, real price, real dividend, real earnings, and PE10 (Shiller CAPE ratio).

2.2 Feature Engineering

Raw prices are non-stationary and highly auto-correlated — poor inputs for clustering. We engineer 8 features that capture the economic state of each month.

Feature Selection Rationale

Feature selection was driven by three explicit criteria, applied before any clustering was run:

- **Stationarity.** K-Means operates on Euclidean distances, which are meaningless on trending series. Raw nominal prices rise monotonically over 142 years, so any distance computed on them reflects the passage of time rather than economic similarity. Every engineered feature

is either a ratio (PE10, Dividend Yield, Earnings Yield), a first difference (Rate Change), a percentage change (Inflation MoM, Real Return MoM), or a rolling return (12m Momentum) — all of which are stationary or near-stationary, ensuring the clustering captures economic state rather than trend.

- Coverage of the economic state space. A single valuation metric would leave entire dimensions of the macroeconomic environment unrepresented. We therefore selected features to span four orthogonal economic dimensions: (i) valuation — PE10, Earnings Yield, Dividend Yield; (ii) monetary environment — Long Interest Rate, Rate Change; (iii) inflation — Inflation MoM; (iv) market dynamics — 12m Momentum, Real Return MoM. These four dimensions map directly to what practitioners use to characterise regimes (Ang & Timmermann, 2011), and our feature correlation matrix confirms they are largely uncorrelated, meaning each adds independent discriminating power.
- Domain grounding. Each feature has an established macroeconomic interpretation and prior use in the regime-detection literature. PE10 (the Shiller CAPE ratio) is the primary valuation signal in Shiller (2000) and is the most widely cited long-horizon return predictor. Twelve-month momentum is a well-documented predictor of regime transitions (Ang & Timmermann, 2011). Long interest rates anchor the monetary dimension in virtually all macro regime models. Inflation MoM captures the supply-side shock component that drove the 1970s High Rate Era cluster. Using only features with clear economic meaning also ensures the resulting clusters are interpretable — a pure statistical feature (e.g., principal component scores) would cluster well but tell us nothing actionable.

Features that were considered and excluded include raw nominal price (non-stationary), raw earnings (collinear with PE10 once price is known), and rolling volatility (captures risk, not regime — added noise in preliminary EDA without improving silhouette). The final 8-feature set was fixed before fitting any model to prevent data snooping.

Feature	Formula / Source	Economic Intuition
PE10 (CAPE)	Shiller dataset	Market valuation level
Long Interest Rate	Shiller dataset	Monetary environment
Dividend Yield	$\text{Real Dividend} / \text{Real Price} \times 100$	Income return; high = cheap market
Earnings Yield	$1 / \text{PE10} \times 100$	Inverse of PE10
Inflation MoM	$\text{CPI pct_change} \times 100$	Monthly inflation signal
Rate Change	$\text{diff}(\text{Long Rate})$	Rate momentum / policy shift
12m Momentum	$(P_t / P_{t-12} - 1) \times 100$	Trend signal
Real Return MoM	$\text{Real Price pct_change} \times 100$	Short-term return / volatility

Table 0. Feature engineering summary. All features are stationary and span four orthogonal economic dimensions.

2.3 Preprocessing

K-Means uses Euclidean distance, so features on different scales would dominate. **StandardScaler** (zero mean, unit variance) is applied so each feature contributes equally to the distance computation. After scaling, all means are ~ 0 and all standard deviations ~ 1 , verified programmatically.

2.4 k Selection — Three-Metric Framework

We test $k = 2$ through 9 and evaluate three complementary metrics:

Metric	What it measures	Target
Inertia (Elbow)	Total within-cluster variance	Elbow / bend point
Silhouette Score	Separation vs cohesion (-1 to 1)	Higher is better
Davies-Bouldin	Avg within/between ratio	Lower is better

Figure 1. k -selection metrics for $k = 2 - 9$. The elbow at $k=4$ combined with a local Davies-Bouldin minimum and stable silhouette guides our choice.

The elbow at $k=4$ is the most prominent inflection point in the inertia curve. Davies-Bouldin reaches a local minimum at $k=4$ and $k=5$ (1.497 vs 1.464), while silhouette peaks at $k=2$ but remains broadly stable through $k=5$. We choose $k=4$ because it aligns with the elbow, offers an economically interpretable four-regime taxonomy (bull, fairly valued, high-rate, bear), and matches the hypothesis framed in the proposal.

2.5 Clustering Algorithms

K-Means (primary): Minimises total within-cluster variance by iteratively assigning points to the nearest centroid and recomputing centroids. Run with $n_init=50$ and $max_iter=500$ to ensure convergence. Formally: given $X \in \mathbb{R}^{1556 \times 8}$, find $f: \mathbb{R}^8 \rightarrow \{0,1,2,3\}$ minimising $J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$.

Agglomerative Clustering: Hierarchical bottom-up approach with Ward linkage — merges clusters that minimise the increase in total within-cluster variance. Makes no centroid assumption and handles non-spherical shapes.

DBSCAN: Density-based algorithm that finds clusters as dense regions separated by sparse space, automatically labelling low-density points as outliers (noise). Parameters: $eps=1.5$, $min_samples=20$.

2.6 PCA for Visualisation

PCA projects the 8-dimensional feature space to 2D for visual inspection. PC1 explains 36.7% of variance and is dominated by valuation features (PE10, Earnings Yield, Dividend Yield). PC2 explains 16.0%, driven by Inflation MoM and Rate Change. Together they capture 52.7% — sufficient to confirm cluster separation visually.

3. Results

3.1 Cluster Profiles

Regime	n (months)	Avg PE10	Avg Rate	Avg 12m Return	Avg Div Yield
0 — Bull / Overvalued	649 (41.7%)	27.3	4.1%	+14.2%	2.8%
1 — Fairly Valued	594 (38.2%)	17.9	3.4%	-0.3%	4.9%

Regime	n (months)	Avg PE10	Avg Rate	Avg 12m Return	Avg Div Yield
2 — High Rate Era	198 (12.7%)	14.3	9.4%	-0.1%	5.6%
3 — Bear / Crash	115 (7.4%)	10.3	4.1%	-20.4%	7.1%

Table 1. Mean profile for each discovered regime. Regimes are ordered by descending PE10 to reflect a valuation hierarchy.

Figure 2. Feature distributions by regime. Each box shows median, IQR, and outliers. The separation in PE10 and 12m Momentum is clear and economically intuitive.

Economic interpretation: Cluster 0 (Bull) is characterised by high PE10 (27.3) and strongly positive 12-month momentum (+14.2%), consistent with the 1990s dot-com era and 2010s bull market. Cluster 1 (Fairly Valued) covers the majority of 20th-century history with moderate valuations and flat returns. Cluster 2 (High Rate Era) has the highest interest rates (9.4%) and captures the 1970s–1980s inflation period. Cluster 3 (Bear/Crash) shows very low PE10 (10.3) and severe negative momentum (–20.4%), aligned with the Great Depression and 1970s stagflation.

3.2 PCA Visualisation & Timeline

Figure 3. Left: cluster assignments in PCA space. Right: PE10 vs Interest Rate coloured by regime — the High Rate Era (orange) separates clearly on the x-axis.

Figure 4. Market Regime Timeline 1881–2023. Top: real S&P 500 price (log scale) coloured by regime. Bottom: regime colour strip showing historical era assignments.

The timeline validates the clustering against known history. Bull regimes (gold) dominate post-1980s. The High Rate Era (orange) concentrates around 1965–1985. Bear regimes (red) align with the Great Depression (1929–1933), 1970s dips, and early 2000s crash. Regimes are persistent — the transition matrix (Section 3.4) confirms this formally.

3.3 Algorithm Comparison

Algorithm	Clusters	Outliers	Silhouette ↑	Davies-Bouldin ↓
K-Means (k=4)	4	0	0.2054	1.5008
Agglomerative (k=4)	4	0	0.1645	2.0918
DBSCAN (eps=1.5)	1	243	N/A (1 cluster)	N/A

Table 2. Algorithm comparison. K-Means achieves the best silhouette (0.2054) and Davies-Bouldin (1.5008). DBSCAN collapses to a single cluster at eps=1.5 — a finding in itself, discussed below.

Figure 5. PCA scatter for all three algorithms. K-Means and Agglomerative produce visually similar partitions; DBSCAN labels 243 points as noise and finds only 1 dense cluster.

Why DBSCAN collapses: Macro data in 8 dimensions is dense and relatively uniformly distributed — there are no clearly separated dense islands of the kind DBSCAN expects. Varying eps from 1.0 to 3.0 consistently yielded either 1 mega-cluster or many tiny ones. This is itself informative: market regimes are not sharply separated in feature space but rather form a continuum — centroid-based methods like K-Means are better suited.

3.4 Cluster Validation — Silhouette & Transition Matrix

Figure 6. Silhouette plot for K-Means k=4. Average silhouette = 0.205, which is reasonable for real-world, high-dimensional macro data with inherent overlap.

A silhouette score of 0.2 may seem modest, but is expected: macroeconomic data has no sharp phase boundaries. Markets transition gradually. The score is consistent with published regime clustering literature on similar macro datasets (Nystrup et al. 2018).

→	Bull	Fairly Valued	High Rate Era	Bear
Bull	0.89	0.08	0.02	0.01
Fairly Valued	0.10	0.86	0.03	0.01
High Rate Era	0.01	0.06	0.93	0.00
Bear	0.02	0.13	0.05	0.80

Table 3. Month-to-month regime transition probability matrix. Strong diagonal confirms market regimes are highly persistent.

The diagonal-dominant transition matrix confirms markets are 'sticky': the High Rate Era has a 93% self-transition probability and average duration of 14.1 months; the Bull regime stays for ~9.4 months on average; Bear crashes are shorter at ~5.0 months. This persistence is practically useful — regime labels can serve as state variables in downstream supervised models without severe look-ahead contamination.

4. Interpretability — Feature Importance

Since K-Means has no built-in feature importance, we assess interpretability via PCA loadings and cluster centroid analysis.

Feature	Importance Signal	Key Evidence
PE10 (CAPE)	#1 — Primary	PC1 loading 0.543; largest centroid gap (10.3 Bear vs 27.3 Bull)
12m Momentum	#2 — Strong	Separates Bull (+14.2%) from Bear (−20.4%) most cleanly
Long Interest Rate	#3 — Defining	Sole driver of High Rate Era cluster (9.4% avg vs 4.5% overall)
Dividend Yield	#4 — Inverse	High in cheap markets: Bear 7.1% vs Bull 2.8%

PE10 and 12m Momentum are the two most discriminating features, consistent with Shiller (2000) and Ang & Timmermann (2011). PCA confirms this: PC1 (36.7% variance) is driven entirely by valuation features (PE10, Earnings Yield, Dividend Yield), while PC2 (16.0%) captures the inflation/rate axis. The PE10 vs Interest Rate scatter (Figure 3) shows all four clusters are meaningfully separated in this reduced view — no cluster is fully contained within another.

5. Discussion & Conclusions

What worked: K-Means with k=4 produced economically coherent and historically interpretable regimes. The three-metric k-selection framework gave a principled choice. Feature engineering from

raw Shiller data was the critical preprocessing step — clustering raw prices produces meaningless results due to non-stationarity.

What didn't work: DBSCAN was not appropriate for this data — macro data forms a continuous manifold without clear density gaps. Agglomerative clustering underperformed K-Means on both metrics (Silhouette 0.1645 vs 0.2054; DB 2.09 vs 1.50), suggesting that the clusters are roughly spherical and K-Means centroid assumptions are valid here.

Limitations: A silhouette of 0.205 is modest. Macro regimes are not crisp — transitions are gradual. The dataset spans 142 years with structural breaks (gold standard, Bretton Woods, NASDAQ era) that a single global clustering cannot fully capture. Hidden Markov Models (HMM) would handle smooth probabilistic transitions better.

What we learned: Feature engineering matters more than algorithm choice for tabular macro data. Domain knowledge (PE10 as valuation, momentum as trend) directly informed which engineered features to include. The DBSCAN failure taught us that market regime boundaries are fuzzy — a key finding for practitioners.

Future work: (1) Semi-supervised extension — use regime labels as features in a supervised return-prediction model. (2) Hidden Markov Model for probabilistic regime transitions with uncertainty quantification. (3) Apply to other asset classes (bonds, commodities) to test cross-asset regime coherence.

6. References

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7. Team Work

Task	Owner	Date
Proposal Presentation	Both	Feb 23
Data Cleaning & Feature Engineering (8 features)	Dusan Barudzija	Mar 1
K-Means — Elbow, Silhouette & Davies-Bouldin	Danish Kumar	Mar 8
Agglomerative & DBSCAN — Algorithm Comparison	Dusan Barudzija	Mar 15

Task	Owner	Date
Cluster Profiling, PCA Visualisation & Timeline	Both	Mar 22
Final Report Draft	Both	Mar 29
Final Presentation & Demo	Both	Early Apr

Dusan led data engineering and algorithm comparison; Danish led K-Means implementation and evaluation. Report writing, cluster profiling, and visualisation were completed collaboratively in shared Jupyter sessions.